

From a credential registry to two composable standards

PR #204 proposed a registry and ANS / Scan discovery. The new drafts separate credentials from DID discovery.

ORIGINAL · PR #204

Registry first

Credential registration with proposed ANS / Scan discovery.

NOW · TWO EARLY DRAFTS

Credentials + DID

Credentials defines the contract and lifecycle. **DID** defines identity and resolution.

PROJECT DELIVERABLE

DSO reference

Build the DSO Credential Registry and DSO DID Registry together. Standards adopters need not run either.

Draft evolution, not a production migration. Both DSO registries are project deliverables, not requirements on standards adopters.

One credential. Registration adds a separate lifecycle.

`Credential` and `RegisteredCredential` are views of the same contract, not competing identities.

INTRINSIC CONTRACT

Credential

Holds the intrinsic credential data and validity. It can exist without a registry.

`validUntil` governs credential validity.

OPTIONAL REGISTRY VIEW

RegisteredCredential

Exposes registration for that credential, when registration is chosen.

`expiresAt` governs registry retention.

One canonical contract; the registered view adds registry metadata. DSO paid retention would extend `expiresAt`, never `validUntil`. Payment and authorization for that extension remain open.

The W3C issuer may be external. The anchor is on Canton.

ISSUER

W3C identifier

The issuer identifier need not be a Canton Party. An external identifier does not become a Canton signatory.

ANCHORING

Canton parties

Anchorers are mandatory Canton parties responsible for anchoring the credential.

HOLDING

Optional holders

Holders are optional Canton parties. The ledger contract remains Canton-specific.

Signatories: anchorers plus holders, deduplicated. Anchoring records a claim; it does not verify the claim. Encoding issuer or holder as a mandatory Canton Party in downstream templates would force migration or a false Party identity.

Who holds it? Who is it about?

Alice holds an ACME licence. Alice is the holder; ACME is the explicit credential subject.

OLD KEY CONVENTION

Subject hidden in a key

`licenseStatus!ACME` identifies ACME by convention; bare `licenseStatus` implies Alice. This conflates claims and subject selection.

DRAFT MODEL

Subject is explicit

Set the credential subject to ACME and use `licenseStatus` as the claim key. Holder Alice remains a separate role.

A compatibility projection for old key prefixes is a **proposal**, not implemented behavior. Do not infer the subject from a claim key in the new model.

Find an endpoint. Verify it separately.

The drafts explore typed, versioned DID Document service entries instead of proposed ANS claim URLs / Scan discovery:

01

DID identifier

Identify the DID and its controller.

02

Registrar

Candidate publisher of DID state.

03

Resolver

Would verify DID state under selected method rules.

04

Service entry

Points to a versioned Credentials API.

A wallet or explorer could resolve a registry DID and call the versioned API without a Scan proxy. DID binding and Party Profiles PR #169 mapping remain open. The selected multi-endpoint resolution POC compares endpoints; it does not deliver BFT.

Issue and renew. Register and retain separately.

Two clocks govern different things; neither requires a transaction merely because time passes.

INTRINSIC LIFECYCLE

Issue and renew

Factory issuance creates a credential. Renewal changes `validUntil` only: credential validity, not registry retention.

REGISTRY LIFECYCLE

Register, extend, archive

Registration is optional. Extension changes `expiresAt` only: registry retention, not credential validity. Archiving is separate.

Selected Daml lifecycle POC: compare these transitions. Offer / Accept, AddHolder, suspension, resumption and revocation remain open, **not delivered**. External off-ledger W3C proof verification is future work, not implemented in Daml.

Two resources. Active and archived reads.

The selected PQS POC tests API reads, not complete ledger choice history.

HTTP RESOURCES

Intrinsic credential

Expose independently of registration; no registry record required.

HTTP RESOURCES

Registered credential

Separate resource; credential data and registration metadata stay distinct.

PQS READS

Active records

Fixed-offset, keyset-paginated reads.

PQS READS

Archived records

Fixed-offset, keyset-paginated reads.

The PQS POC does **not** promise complete choice-event replay. Reads and lifecycle are separate concerns.

TAKEAWAY

Separate the contract, the registry and discovery.

Build the DSO Credential Registry and DID Registry as a reference implementation of two composable early drafts.

CREDENTIAL

Intrinsic truth

Explicit subject and roles; validity belongs to the credential.

REGISTRATION

Optional retention

Registry metadata and retention have their own lifecycle.

DID + API

Discover, then verify

Discovery points to an API; trust and read guarantees need separate decisions.

Selected POCs test lifecycle, PQS reads and multi-endpoint DID resolution. None alone delivers combined DSO BFT or payment.